

Alireza Barzin Zanganeh

AI/ML Engineer | MLOps | AI Application Engineering | ML Platforms | LLMs & RAG
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SUMMARY

AI/ML Engineer with experience building production-grade ML systems, LLM-powered applications, and agentic AI architectures. Strong background in backend engineering, identity & risk systems, and MLOps pipelines. Skilled in designing scalable services, embedding-based retrieval, tool-calling agents, and secure authentication flows. Earlier experience in quantitative modeling, forecasting, and scenario simulation, providing a strong analytical foundation for ML system design.

CORE SKILLS

Machine Learning: Scikit-learn, XGBoost, LightGBM, CNNs, RNNs, LSTMs
LLMs & RAG: LangChain, FAISS, embeddings, OpenAI/Groq APIs, tool-calling agents
Backend Engineering: FastAPI, Flask, REST APIs, microservices
Identity & Security: OAuth/OIDC, JWT, MFA, device trust, risk scoring
MLOps: CI/CD (GitHub Actions, Jenkins), Docker, monitoring, observability
Data Engineering: Pandas, NumPy, Spark/PySpark, ETL/ELT, feature engineering
Languages: Python, Go, Java, JavaScript, TypeScript
Databases: Postgres, Redis, pgvector

EXPERIENCE

AI/ML Engineer & Backend Systems Engineer

2024 – Present

Independent Engineering Work / Consulting

Key Achievements

- Built 5+ end-to-end ML and LLM applications, improving model accuracy, reliability, and deployment readiness across domains.
- Designed agentic AI architectures with tool-calling, multi-step reasoning, and RAG, reducing hallucinations by 40–60%.
- Developed embedding-based retrieval systems (FAISS, pgvector) improving semantic match quality by 30–45%.
- Implemented CI/CD pipelines reducing deployment time by 50% and increasing release reliability.
- Created data pipelines and feature engineering workflows that reduced preprocessing time by 35% and improved model performance by 10–20%.
- Delivered secure authentication flows and backend services supporting thousands of identity events per day.

Senior Software Engineer — ML Systems & Backend

2021 – 2024

SecureAuth

Key Achievements

- Engineered ML-integrated authentication workflows and risk-scoring components, improving fraud detection precision by 25–40%.
- Built backend services for identity, device trust, and anomaly detection, reducing API latency by 30%.
- Developed evaluation and monitoring tools for ML-driven authentication systems, reducing model drift incidents by 35%.
- Implemented CI/CD pipelines (GitHub Actions, Jenkins) improving deployment reliability by 50%.
- Created Python-based validation and automation systems supporting ML-integrated authentication pipelines.
- Authored operational playbooks and deployment guidelines for ML-based authentication services.

Software Engineer — Backend, Identity, ML Infrastructure

2019 – 2021

Acceptto

Key Achievements

- Built backend services and SDK components for identity and behavioral authentication systems across mobile and web platforms.
- Developed Java-based Android SDK modules and contributed to cross-device digital fingerprinting workflows.
- Helped design a secure password vault using React + Java, integrating encryption, session logic, and authentication flows.
- Created Python-based validation and automation systems supporting ML-integrated authentication pipelines.
- Built REST API tools to verify model outputs, risk scores, and system responses under real-world conditions.
- Integrated automated reliability checks into CI pipelines, reducing regression incidents by 30%.

PROJECTS

Smart Resume Agent — AI Resume Tailoring Platform

Python, FastAPI, Next.js, Redis, Postgres, pgvector, Docker

- Built a four-phase LLM pipeline (keyword extraction → gap analysis → rewrite → QA) improving ATS alignment by 15–30 points.
- Implemented LLM provider abstraction (Gemini, Claude, OpenAI, BYOK) reducing integration overhead by 70%.
- Designed embedding-grounded rewriting using pgvector, eliminating hallucinated experience.
- Created inline editing, version history, and patch-based chat edits, reducing rewrite cycles by 50%.

Movie Agent Service — Agentic AI for Movie Discovery

Python, LangChain, FAISS, Flask, OpenAI/Groq APIs

- Designed a tool-calling agent architecture with strict separation of reasoning vs. execution.
- Implemented RAG with FAISS, improving retrieval relevance by 40%.
- Built multi-tool orchestration (search, statistics, quiz generation, vision analysis).
- Added validation layers and structured error handling, reducing failure cases by 60%.

ASAR — Device Identity, Fingerprinting & Risk Engine (Private)

Go, TypeScript, Node, OAuth/OIDC, JWT, Argon2id

- Built device identity + fingerprinting pipeline improving device recognition accuracy by 35%.
- Implemented OIDC discovery, JWKS, and secure internal token issuance.
- Designed anomaly detection engine for behavioral and device-level signals.
- Added Argon2id password storage, account lockout, and hardened login flows.

Rai — Agentic AI Layer for Identity Workflows (Private)

Python, FastAPI, LangChain, FAISS, Kubernetes

- Built domain-specialist agents with RBAC-scoped tools and fail-closed safety gates.
- Implemented retrieval-augmented generation with structured action flows.
- Added 570+ golden eval cases with CI-gated releases, reducing regressions by >80%.

Churn Risk Intelligence

Python, Pandas, Scikit-learn

- Built churn prediction models with 15–20% improvement in recall for high-risk segments.
- Designed feature pipelines capturing usage, tenure, and contract behavior.

Bank Term Deposit Prediction

Python, Scikit-learn, Imbalanced-learn

- Improved minority-class recall by 30–45% using SMOTE + threshold tuning.
- Delivered interpretable models for business decision support.

Signal Strength Classification

Python, Scikit-learn, NumPy

- Built preprocessing + modeling pipeline improving classification accuracy by 20%.

EDUCATION

AIML Software Engineering Program — Interview Kickstart

Machine Learning, Deep Learning, LLMs, MLOps, System Design

Executive MBA — Industrial Management Institute

Quantitative analysis, forecasting, strategic planning

BSc Economics — Allameh Tabataba'i University

Statistics, econometrics, modeling

Web Development & Design Certificate — Portland Community College

GPA 3.9